

Lecture 7 - Local Average Treatment Effects

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Questions

- When extracting LATE, I see that we mainly are focused on the LATE on the compliers. Are we expected/can we find the LATE for the never-takers and always-takers and would those LATE estimates be causal? (I assume compliers LATE is the only causal and relevant effect to derive but worth asking)

- Continue our discussion of causal effects under the potential outcomes framework $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$
- We have shown that the average treatment effect $E[\beta_i] = E[Y_i(1) - Y_i(0)]$ can be point identified in the following cases:
 - Randomized control trials, where $(Y_i(0), Y_i(1)) \perp D_i$
 - Selection on observables: $(Y_i(0), Y_i(1)) \perp D_i | X_i$ (can estimate CATE first or estimate propensity scores, and form $\frac{D_i Y_i}{\hat{p}_i}, \frac{(1-D_i) Y_i}{1-\hat{p}_i}$)
 - IV method if RCT has incomplete compliance: constant treatment effect $\beta_1 \equiv Y_i(1) - Y_i(0)$ can be identified via an instrument Z_i for D_i :

$$\hat{\beta}_1^{IV} = \frac{\bar{Y}^1 - \bar{Y}^0}{\bar{D}^1 - \bar{D}^0} \xrightarrow{p} \beta_1 \text{ constant TE} \sim \text{ATE}$$

where $\bar{Y}^1 = \frac{1}{N_1} \sum_{i: Z_i=1} y_i$ (sample mean among those randomly assigned to $Z_i = 1$).

- This lecture we continue to work with the general framework allowing for treatment effect heterogeneity:

$$\beta_{1i} := Y_i(1) - Y_i(0)$$

- Can we identify ATE $E[\beta_{1i}]$ using the IV method, or the ATE for some sub-population? What does $\hat{\beta}_1^{IV}$ converge to?
- Local average treatment effects (IV-LATE)
 - ATE for compliers (but not always takers/never takers/defiers)
 - characterize compliers, in comparison with other groups.
 - Application: Self-Sufficiency Project, C-section

Heterogeneous Treatment Effects

- Causal model of interest:

$$\begin{aligned}Y_i &= D_i Y_i(1) + (1 - D_i) Y_i(0) \\&= Y_i(0) + D_i \times (Y_i(1) - Y_i(0)) \\&= Y_i(0) + D_i \times \beta_{1i}\end{aligned}\tag{1}$$

- Define ATE $\beta_1 = E[\beta_{1i}]$, and $\beta_0 := E[Y_i(0)]$, (1) can be rewritten as:

$$\begin{aligned}Y_i &= \beta_0 + \beta_1 D_i + u_i \\ \text{where } u_i &= Y_i(0) - \beta_0 + (\beta_{1i} - \beta_1) D_i\end{aligned}$$

the residual u_i is clearly correlated with D_i unless $D_i \perp (Y_i(0), Y_i(1))$ (as in a perfect RCT).

- If you estimate a population regression $Y_i = b_0 + b_1 D_i + \epsilon_i$, what is b_1 ?

Instrument Variable Approach

- Suppose we have a valid instrument for treatment D_i : Z_i predicts D_i and $E[Z_i u_i] = 0$. Turns out the IV approach can help us get consistent estimates for the average treatment effects in some **sub-population (compliers)**.
- Let's look at an *experiment to study the effect of smoking by mothers on birthweight*.
 - Why not a RCT? Can't command mothers-to-be to smoke or not. So: conduct an “encouragement design” experiment among pregnant women who smoke and want to quit.
 - Z_i = **treatment** status = 1 if given “treatment” (counseling)
 - D_i = dummy for “desist” or stop smoking (1=desist, 0=keep smoking)

Instrument Variable Approach

- Y_i = weight of newly born infant

$$Y_i = \beta_0 + \beta_1 D_i + u_i \quad (\text{causal model}) \quad (2)$$

$$D_i = \pi_0 + \pi_1 Z_i + \eta_i \quad (\text{1st stage})$$

$$Y_i = \underbrace{\beta_0 + \beta_1 \pi_0}_{\delta_0} + \underbrace{\beta_1 \pi_1}_{\delta_1} Z_i + \underbrace{(u_i + \beta_1 \eta_i)}_{\nu_i}$$

by exclusion restriction and the first-stage regression, $Z_i \perp (u_i, \eta_i)$.

Therefore, we can consistently estimate the reduced-form regression:

$$Y_i = \delta_0 + \delta_1 Z_i + \nu_i$$

$$\delta_1 = \beta_1 \pi_1$$

- This is the “dummy-dummy” case (both D_i and Z_i are binary).
What are first stage and reduced form coefficients in sample?

$$\begin{aligned} \hat{\pi}_1 &= \overline{D^1} - \overline{D^0} \\ \hat{\delta}_1 &= \overline{Y^1} - \overline{Y^0} \\ \Rightarrow \hat{\beta}_1^{iv} &= \frac{\overline{Y^1} - \overline{Y^0}}{\overline{D^1} - \overline{D^0}} \end{aligned} \quad (3)$$

Instrument Variable Approach

- We are dividing the difference in birthweights (between treatment and control groups) by the difference in the fraction of mothers who quit.
- What does $\hat{\beta}_1^{iv}$ represent? From (3), note $\hat{\beta}_1^{iv}$ is driven by the gains in birth outcomes for the **moms who quit smoking because of the encouragement**.
- Let's formalize this interpretation!

Instrument Variable in Potential Outcome Framework

- Let's think about how people behave and how they can respond to “treatment” ($Z_i = 1$, e.g. receive counseling).
- Let's introduce 2 new random variables in the potential outcome framework – that describe types of people and how they react: $(D_i(0), D_i(1))$
 - $D_i(0)$ = indicator of desist if person i is assigned to $z = 0$
 - $D_i(1)$ = indicator of desist if person i is assigned to $z = 1$
- Now we can characterize/classify individuals based on their **compliance type** $(D_i(0), D_i(1))$:
 - $D_i(0) = 0, D_i(1) = 0$ – “never takers” - never desist
 - $D_i(0) = 1, D_i(1) = 1$ – “always takers” - always desist
 - $D_i(0) = 0, D_i(1) = 1$ – “compliers” - desist if treated
 - $D_i(0) = 1, D_i(1) = 0$ – “defiers” - desist if untreated, not if treated.
- **(Random assignment)** All 4 groups should be equally likely in T and C groups.

$$(D_i(0), D_i(1)) \perp Z_i$$

which implies $\Rightarrow Pr(\text{Compliers}|Z_i) = Pr(\text{Compliers})$, and also for AT/NT/Defiers.

$$D_i = D_i(z_i)$$

 $-z$
 $D_i(0)$
 $D_i(1)$

Never-Taker

0

0

Always-Taker

1

1

Compliers

0

1

Defiers

1

0

no defiers if
 $D_i(1) \geq D_i(0)$

randomly assigned
to treatment
(receiving counseling)

not assigned

IV - First Stage

- Who is still smoking when kid is born?
 - in Control-group ($Z_i = 0$) : smokers = $NT's + Comp's$
 - in Treated-group ($Z_i = 1$) : smokers = $NT's$ (if no defiers)
- So: if no defiers, difference in fraction of smokers = fraction of **Compliers!** The first-stage under the random assignment assumption:

$$D_i = \pi_0 + \pi_1 Z_i + \eta_i$$

$$\pi_1 = E[D_i | Z_i = 1] - E[D_i | Z_i = 0]$$

$$= E[D_i(1) | Z_i = 1] - E[D_i(0) | Z_i = 0]$$

$$= Pr(D_i(1) = 1 | Z_i = 1) - Pr(D_i(0) = 1 | Z_i = 0)$$

$$= Pr(D_i(1) = 1) - Pr(D_i(0) = 1) \text{ by random assignment}$$

$$= Pr(AT + Comp) - Pr(AT + Defiers)$$

this dropped
by random
assignment

as $(D_i(0), D_i(1))$ is
independent of (Z_i)

The assumption $Pr(Defiers) = 0$ is called the **monotonicity assumption**:

$$\forall i : D_i(1) \geq D_i(0)$$

IV - First Stage

So
flip?

	$D_i(0)$	$D_i(1)$
AT	0	0
NT	1	1
Compliers	1	0
Def	0	1

- note: if we flip the definition of $D_i = \text{smoking}$, this assumption is $\forall i : D_i(1) \leq D_i(0)$. There is no defier with $D_i(1) = 1$ while $D_i(0) = 0$.
- In summary, under the random assignment + monotonicity assumptions:

$$\pi_1 = Pr(Comp)$$

the first-stage coefficient is the fraction of compliers (characterized by $D_i(1) = 1, D_i(0) = 0$) in the population.

IV - 2nd Stage $Y_i(d, z)$

- Clearly the difference in birth weights between the $z = 0$ and $z = 1$ groups depends on the changes experienced by the compliers as a result of the intervention in the $z = 1$ environment. To formalize that we revise the *potential outcomes* for y :

$$Y_i(d, z) = \text{outcome for } i \text{ if } D_i = d \text{ and } Z_i = z$$

Now if z has no direct effect on the outcome:

$$\begin{aligned} Y_i(0, 0) &= Y_i(0, 1) = Y_i(0) \text{ the outcome if } D_i = 0 \\ Y_i(1, 0) &= Y_i(1, 1) = Y_i(1) \text{ the outcome if } D_i = 1 \end{aligned}$$

Handwritten notes:
Blue arrow pointing to $Y_i(0)$: "this is when $D_i = 0$ "
Blue arrow pointing to $Y_i(1)$: "this is when $D_i = 1$ "

where $Y_i(d)$ is the potential outcome when unit i has $D_i = d$ (as before).

- We often write the **exclusion restriction** assumption as:

$$Y_i(d, z) = Y_i(d)$$

IV - 2nd Stage $Y_i(d, z)$

- Recap: Part of what it means to be a “good instrument” is that - z only works through its effect on D . We will assume this “exclusion restriction” is true. When we write

$$Y_i = \beta_0 + \beta_1 D_i + u_i \quad (\text{causal model})$$

$$D_i = \pi_0 + \pi_1 Z_i + \eta_i \quad (\text{1st stage})$$

we are *building in the exclusion restriction*. Note that u_i in causal model here w/ this set up will be orthogonal to Z_i .

- Z_i does not appear in the structural model
- since $E[Z_i u_i] = 0$, changes in z_i do not affect u_i . Thus the ONLY way for Z_i to affect Y_i is via the effect on X_i . $\rightarrow E(Z_i X_i) \neq 0$ and $E(z_1 \dots) = 0$

$$\begin{array}{ccccccc} (Y_i(0), & Y_i(1), & D_i(0), & D_i(1)) & \perp\!\!\!\perp & Z_i \\ \downarrow & \downarrow & \downarrow & \downarrow & & \\ d=0 & d=1 & z=0 & z=1 & & \end{array}$$

IV - 2nd Stage: Realized Y_i by group

$D_i(0)$	$D_i(1)$
$z_i = 0$	$z_i = 1$

AT
N
Compliers
Defiers

→ see
slide

The realized outcome for unit i can be expressed as:

$$Y_i = Y_i(D_i) = Y_i(D_i(Z_i)) \quad (4)$$

Let's break it down by group:

- Never takers ($D_i(0) = 0, D_i(1) = 0$): $D_i = 0$ regardless of $z_i \Rightarrow Y_i = Y_i(0)$ for either z_i .
- Compliers ($D_i(0) = 0, D_i(1) = 1$): observe $Y_i = Y_i(0)$ if $z_i = 0$, observe $Y_i = Y_i(1)$ if $z_i = 1$
- Always takers ($D_i(0) = 1, D_i(1) = 1$): $D_i = 1$ regardless of $z_i \Rightarrow Y_i = Y_i(1)$ for either z_i
- Defiers ($D_i(0) = 1, D_i(1) = 0$): observe $Y_i = Y_i(1)$ if $z_i = 0$, $D_i = 0$ if $z_i = 1 \Rightarrow$ observe $Y_i = Y_i(0)$ if $z_i = 1$

g	$Z = 0$		$Z = 1$	
	$D_i(0)$	$Y_i(D_i(0))$	$D_i(1)$	$Y_i(D_i(1))$
NT	0	$Y_i(0)$	0	$Y_i(0)$
AT	1	$Y_i(1)$	1	$Y_i(1)$
Compilers	0	$Y_i(0)$	1	$Y_i(1)$
Defiers	1	$Y_i(1)$	0	$Y_i(0)$
Realized Y_i	$E[Y_i Z_i = 0] = E[Y_i(0) NT] \cdot Pr(NT) \\ + E[Y_i(0) Comp] \cdot Pr(Comp) \\ + E[Y_i(1) AT] \cdot Pr(AT) \\ + E[Y_i(1) Def] \cdot Pr(Def)$		$E[Y_i Z_i = 1] = E[Y_i(0) NT] \cdot Pr(NT) \\ + E[Y_i(1) Comp] \cdot Pr(Comp) \\ + E[Y_i(1) AT] \cdot Pr(AT) \\ + E[Y_i(0) Def] \cdot Pr(Def)$	

IV - 2nd Stage: Realized Y_i by group

$$Y_i = \delta_0 + \delta_1 Z_i + V_i \quad \begin{cases} E[V_i] = 0 \\ E[V_i Z_i] = 0 \end{cases} \begin{matrix} \text{orthogonality} \\ \text{assumption} \end{matrix} \Rightarrow \begin{cases} \delta_0 = E[Y_i | Z_i = 0] \\ \delta_1 = E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0] \end{cases}$$

Now let's think about the mean outcomes we observe in the $z = 0$ world:

$$\begin{aligned} E[Y_i | Z_i = 0] &= E[Y_i(D_i(0)) | Z_i = 0] \text{ by exclusion restriction } Y_i = Y_i(D_i) \\ &= E_g[E[Y_i(D_i(0)) | Z_i = 0, g]] \text{ by the law of iterated exp.} \\ &= \sum_{g \in \{NT, AT, Comp, Def\}} E[Y_i(D_i(0)) | g, Z_i = 0] \times Pr(g | Z_i = 0) \\ &\downarrow \text{apply LIE} \quad \quad \quad \downarrow \\ &= E[Y_i(0) | NT] \times Pr(NT) \\ &\quad + E[Y_i(0) | Comp] \times Pr(Comp) \\ &\quad + E[Y_i(1) | AT] \times Pr(AT) \\ &\quad + E[Y_i(1) | Def] \times Pr(Def) \end{aligned}$$

notice now everything is exempt of Z_i

the last equation comes from the random assignment assumption $(D_i(1), D_i(0)) \perp Z_i$. So fractions of the various groups (e.g., $Pr(NT)$, etc) and expected potential outcomes (e.g., $E[Y_i(1) | AT]$, etc) do not depend on $z=0$ or 1 status.

IV - 2nd Stage: Realized Y_i by group

$$Y_i(D_i(1))$$

Likewise in the $z = 1$ world: *use the*

$$\begin{aligned} E[Y_i|Z_i = 1] &= E[Y_i(0)|NT] \times Pr(NT) \\ &\quad + E[Y_i(1)|Comp] \times Pr(Comp) \\ &\quad + E[Y_i(1)|AT] \times Pr(AT) \\ &\quad + E[Y_i(0)|Def] \times Pr(Def) \end{aligned}$$

If there are no defiers (under monotonicity assumption $D_i(1) \geq D_i(0)$):

$$\begin{aligned} E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] &= E[Y_i(1) - Y_i(0)|Comp] \times Pr(Comp) \\ &\quad + \underbrace{E[Y_i(0) - Y_i(1)|Defiers]}_{=0} \times Pr(Def) \end{aligned}$$

= 0 even without
the monotonicity
assumption!

IV - 2nd Stage: Realized Y_i by group

★ Make sure

Thus the population regression version of the reduced form model yields

$$\delta_1 = E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0] = E[Y_i(1) - Y_i(0)|Comp] \times Pr(Comp)$$

and the population regression version of the 1st stage model yields

$$\pi_1 = E[D_i|Z_i = 1] - E[D_i|Z_i = 0] = Pr(Comp)$$

So

$$\frac{\delta_1}{\pi_1} = E[Y_i(1) - Y_i(0) | Comp]$$

Under our assumptions, the ratio of the reduced form effect δ_1 to the 1st stage effect π_1 is the average treatment effect for compliers, which is called the local average treatment effects (local to compliers with $D_i(1) > D_i(0)$).

IV-LATE: Summary

Let's summarize the assumptions we have made in a IV-LATE framework, where we can classify individuals into four groups based on

$$(D_i(0), D_i(1)) = \begin{cases} (0, 0) & \text{never takers} \\ (0, 1) & \text{compliers} \\ (1, 0) & \text{defiers} \\ (1, 1) & \text{always takers} \end{cases}$$

- 1 **Random assignment:** treatment Z_i is randomly assigned. All four groups are equally likely to be in treated or in control.

$$(D_i(0), D_i(1)) \perp Z_i$$

- 2 **Exclusion Restriction:** (instrument) Z_i matters for outcome Y_i only through D_i

$$Y_i(d, z) = Y_i(d)$$

- Note 1 & 2 implies the joint distribution $(D_i(0), D_i(1), Y_i(0), Y_i(1))$ is the same whether $Z_i = 0$ or $Z_i = 1$.

- 3 **Monotonicity:** $D_i(1) \geq D_i(0)$ for all units. In other words, there are no defiers.

IV-LATE: Summary

Under the assumptions above, the first-stage regression gives us:

$$D_i = \pi_0 + \pi_1 Z_i + \eta_i$$

$$\pi_1 = E[D_i|Z_i = 1] - E[D_i|Z_i = 0] = Pr(Comp)$$

and the reduced-form regression:

$$Y_i = \delta_0 + \delta_1 Z_i + \nu_i$$

$$\delta_1 = E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]$$

$$= E[Y_i(1) - Y_i(0)|Comp] \times Pr(Comp)$$

therefore the ratio, which is the population “IV”:

$$\beta_1^{IV} = \frac{\delta_1}{\pi_1} = E[Y_i(1) - Y_i(0)|Comp]$$

The IV estimate provides an estimate of the “gain” in Y that compliers obtain as a result of being in the $z = 1$ setting and getting $D_i = 1$ instead of $D_i = 0$.

- Take a closer look at the original model (heterogeneous treatment effects):

$$Y_i = \beta_{0i} + \beta_{1i}D_i = \beta_0 + \beta_1D_i + u_i$$

$\uparrow$ β_{0i} $\uparrow$ β_{1i}
 $Y_i(0)$ $Y_i(1) - Y_i(0)$

in which

$$\beta_0 = E[Y_i(0)]$$

$$\text{ATE} \sim \beta_1 = E[Y_i(1) - Y_i(0)] = E[\beta_{1i}]$$

$$u_i = Y_i(0) - \beta_0 + D_i \times (\beta_{1i} - \beta_1)$$

D_i is correlated with u_i if $\beta_{1i} \neq \beta_1$ (heterog treatment effects) or $\text{cov}(D_i, Y_i(0)) \neq 0$ (selection into treatment by $Y_i(0)$).

- IV estimator: *who would change their behavior if randomly assigned to treatment and takes it vs control*

$$\hat{\beta}_1^{IV} \xrightarrow{P} E[Y_i(1) - Y_i(0) | \text{Comp}] \text{ LATE}$$

which is different from ATE β_1 for the entire population.

- When does LATE = ATE?

- ④ (Boring case) Constant treatment effect: $Y_i(1) - Y_i(0) \equiv \beta_1$, we showed in the previous lecture that IV yields a consistent estimate of the constant TE.
- ② Homogeneous TE across sub-populations:

$$\begin{aligned} E[Y_i(1) - Y_i(0)|AT] &= E[Y_i(1) - Y_i(0)|NT] \\ &= E[Y_i(1) - Y_i(0)|Comp] \\ \Rightarrow E[Y_i(1) - Y_i(0)|Comp] &= E[Y_i(1) - Y_i(0)] \end{aligned}$$

Extrapolation

- Is it plausible $E[Y_i(1) - Y_i(0)|Comp] = E[Y_i(1) - Y_i(0)|AT] = E[Y_i(1) - Y_i(0)|NT]$?
- From the data, we can estimate $E[Y_i(0)|NT]$ and $E[Y_i(1)|AT]$.
Note:

①
if given
monotonicity
then we
can ignore
defiers

② Never takers *

$$E[Y_i|D_i = 0, Z_i = 1] = E[Y_i(0)|NT, Z = 1] = E[Y_i(0)|NT] *$$

$$E[Y_i|D_i = 1, Z_i = 0] = E[Y_i(1)|AT, Z = 0] = E[Y_i(1)|AT] **$$

③ Always takers **

and also $E[Y_i(0)|Comp]$, $E[Y_i(1)|Comp]$:

$$E[Y_i|\underbrace{D_i = 0, Z_i = 0}_{\text{never takers and compliers}}] = \frac{Pr(NT)}{Pr(NT) + Pr(Comp)} \times E[Y_i(0)|NT] + \frac{Pr(Comp)}{Pr(NT) + Pr(Comp)} \times E[Y_i(0)|Comp]$$

Question: how do you estimate $Pr(NT)$ or $Pr(AT)$? ds. al ready



How do we estimate $Pr(NT)$ or $Pr(AT)$?

* look back at our table; monotonicity $\Rightarrow$ ignore defiers

	$Z=0$		$Z=1$	
g	$D_i(0)$	$Y_i(D_i=0)$	$D_i(1)$	$Y_i(D_i=1)$
NT	0	$Y_i(0)$	0	$Y_i(0)$
AT	1	$Y_i(1)$	1	$Y_i(1)$
compliers	0	$Y_i(0)$	1	$Y_i(1)$

$$E[D_i | Z=0] = Pr(AT)$$

$$E[D_i | Z=1] = Pr(AT) + Pr(C)$$

- Comparing $E[Y_i(0)|NT]$ with $E[Y_i(0)|Comp]$, and $E[Y_i(1)|AT]$ with $E[Y_i(1)|Comp]$, if they are substantially different in levels, then it is much less plausible the ATE for compliers is similar to the average effects for other compliance types.

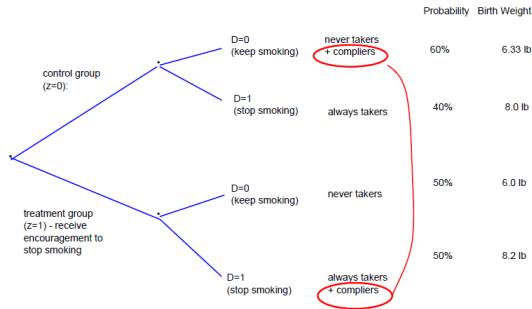
Example: Smoking

Let's revisit the *experiment to study the effect of smoking by mothers on birthweight* (Y_i). $Z_i = 1$ if given "treatment" (counseling). D_i = dummy for "desist" or stop smoking (1=desist, 0=keep smoking)

- First stage: $\hat{\pi}_1 = \bar{D}^1 - \bar{D}^0 = 0.5 - 0.4 = Pr(Comp)$

- Second stage:

$$\hat{\delta}_1 = \bar{Y}^1 - \bar{Y}^0 = (6 * 0.5 + 8.2 * 0.5) - (6.33 * 0.6 + 0.40 * 8) \approx 0.10$$



$$E[D | z=1] - E[D | z=0] = 0.1 \rightarrow Pr(Complier) = .1$$

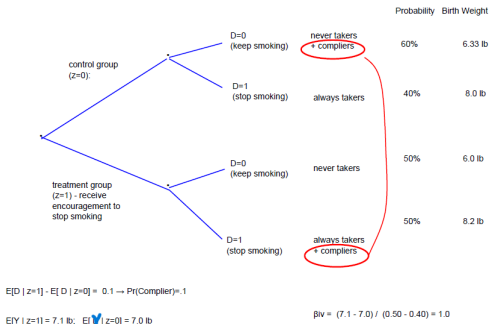
$$E[Y | z=1] = 7.1 \text{ lb}; \quad E[Y | z=0] = 7.0 \text{ lb}$$

$$\beta_{IV} = (7.1 - 7.0) / (0.50 - 0.40) = 1.0$$

Example: Smoking

Is it plausible the ATE for AT or NT are similar to LATE?

- $E[Y_i(1)|AT]$ is approximated by the mean when $D = 0, Z = 1$: 8 lbs
- $E[Y_i|Z_i = 1, D_i = 1] = \frac{Pr(AT)}{Pr(AT)+Pr(Comp)} \times E[Y_i(1)|AT] + \frac{Pr(Comp)}{Pr(AT)+Pr(Comp)} \times E[Y_i(1)|Comp] \Rightarrow (8.2 - \frac{0.4}{0.5} * 8)/0.2 = 9$ is a consistent estimate of $E[Y_i(1)|Comp]$
- $E[Y_i(0)|Comp] = E[Y_i(1)|Comp] - \beta_1^{IV}$: about 8 lbs



Characterize Compliers vs. AT, NT

- Let X_i denote some characteristics for unit i . Another way to see if make sense to extrapolate LATE to other compliance types is to compare observable characteristics of compliers vs. others. It's helpful to know how compliers as a subpopulation are compared with the overall population.
- We can estimate the following from the data:

$$E[X_i|D_i = 0, Z_i = 1] = E[X_i|NT]$$

$$E[X_i|D_i = 0, Z_i = 0] = E[X_i|Comp] \times \frac{Pr(Comp)}{Pr(Comp) + Pr(NT)} \\ + E[X_i|NT] \times \frac{Pr(NT)}{Pr(Comp) + Pr(NT)}$$

$$E[X_i|D_i = 1, Z_i = 0] = E[X_i|AT]$$

$$E[X_i|D_i = 1, Z_i = 1] = \dots$$

You could calculate the mean of X by (D, Z) and work out $\bar{X}^{Comp}$.

Characterize Compliers vs. AT, NT

- But there is an easier way! Define outcome $Y_i = X_i D_i$. Consider this *goofy* 2SLS model:

$$\text{Causal Model: } X_i D_i = a + b D_i + e_i$$

$$\text{First stage: } D_i = \pi_0 + \pi_1 Z_i + \eta_i \quad (5)$$

which yields a reduced-form regression of $X_i D_i$ on instrument Z_i

$$X_i D_i = \delta_0 + \delta_1 Z_i + \nu_i$$

$$\delta_1 = E[X_i D_i | Z_i = 1] - E[X_i D_i | Z_i = 0]$$

$$= E_g[E[X_i D_i(1) | Z_i = 1, g]] - E_g[E[X_i D_i(0) | Z_i = 0, g]]$$

- Under random assignment, Z_i is independent of X_i . What does it imply?

Characterize Compliers vs. AT, NT

- Also note $X_i D_i(1) = X_i$ if $D_i(1) = 1$, 0 otherwise, and $X_i D_i(0) = X_i$ for always takers.

$$\begin{aligned} E_g[E[X_i D_i(0) | Z_i = 0, g]] &= E_g[E[X_i D_i(0) | g]] \\ &= Pr(AT) \times E[X_i | AT] \\ E_g[E[X_i D_i(1) | Z_i = 1, g]] &= E_g[E[X_i D_i(1) | g]] \\ &= Pr(AT) \times E[X_i | AT] \\ &\quad + Pr(Comp) \times E[X_i | Comp] \end{aligned}$$

Therefore:

$$\begin{aligned} \delta_1 &= E[X_i | Comp] \times Pr(Comp) \\ b^{IV} &= \frac{\delta_1}{\pi_1} = E[X | Comp] \end{aligned}$$

we have got the mean characteristics of compliers!

Application: $(Y_i(0), Y_i(1))$ as Characteristics

- Potential outcomes of each unit i can be thought of as characteristics X_i . We can apply/adjust the goofy model above to estimate $E[Y_i(0)|Comp]$, $E[Y_i(1)|Comp]$.
- Note $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0) \sim$ we observe $Y_i(1)$ among units with $D_i = 1$, and $Y_i(0)$ among $D_i = 0$.
- First, construct outcome

$$Y_i(1) \times D_i = \begin{cases} Y_i & D_i = 1 \\ 0 & D_i = 0 \end{cases}$$

and consider “goofy” 2SLS:

$$\text{Causal Model: } Y_i(1)D_i = a_1 + b_1 D_i + \xi_i$$

$$\text{First stage: } D_i = \pi_0 + \pi_1 Z_i + \eta_i \quad (6)$$

Application: $(Y_i(0), Y_i(1))$ as Characteristics

Reduced-form $Y_i(1)D_i = \delta_0 + \delta_1 Z_i + \nu_i$ can be consistently estimated, in which

$$\begin{aligned}\delta_1 &= E[Y_i(1)D_i|Z_i = 1] - E[Y_i(1)D_i|Z_i = 0] \\ &= E[Y_i(1)|Comp] \times Pr(Comp) \\ b_1^{IV} &= E[Y_i(1)|Comp]\end{aligned}$$

Application: $(Y_i(0), Y_i(1))$ as Characteristics

- Second, construct outcome

$$Y_i(0) \times (1 - D_i) = \begin{cases} 0 & D_i = 1 \\ Y_i(0) & D_i = 0 \end{cases}$$

and revise the goofy 2SLS:

$$\begin{aligned} \text{Causal Model: } Y_i(0)(1 - D_i) &= a_0 + b_0 D_i + \xi_i \\ \text{First stage: } D_i &= \pi_0 + \pi_1 Z_i + \eta_i \end{aligned} \quad (7)$$

the reduced-form coef. on Z_i becomes:

$$\begin{aligned} \delta_1 &= E[Y_i(0)(1 - D_i)|Z_i = 1] - E[Y_i(0)(1 - D_i)|Z_i = 0] \\ &= -E[Y_i(0)|Comp] \times Pr(Comp) \\ b_0^{IV} &= -E[Y_i(0)|Comp] \end{aligned}$$

Application: $(Y_i(0), Y_i(1))$ as Characteristics

- The IV estimates b_1^{IV}, b_0^{IV} are linked to LATE:

$$E[Y_i(1) - Y_i(0) | Comp] = b_1^{IV} + b_0^{IV}$$

Is this surprising?

$$Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$$

we estimate the same first stage, 3 reduced form on **the same sample**, where the dependent variables $Y_i(1)D_i + Y_i(0)(1 - D_i)$ add up to Y_i :

$$Y_i = \delta_0 + \delta_1 z_i + \nu_i$$

$$Y_i(1)D_i = \delta_0^{(1)} + \delta_1^{(1)} z_i + \nu_i^{(1)}$$

$$Y_i(0)(1 - D_i) = \delta_0^{(0)} + \delta_1^{(0)} z_i + \nu_i^{(0)}$$

$$Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$$

$$= \left(\delta_0^{(1)} + \delta_0^{(0)} \right) + \left(\delta_1^{(1)} + \delta_1^{(0)} \right) z_i + \left(\nu_i^{(1)} + \nu_i^{(0)} \right)$$

Application: $(Y_i(0), Y_i(1))$ as Characteristics

with $(\delta_1^{(1)} + \delta_1^{(0)}) = \delta_1 = E[Y_i(1) - Y_i(0) | Comp] \times Pr(Comp)$ as the instrument $z_i \perp (\nu_i^{(1)} + \nu_i^{(0)})$ (orthogonal).

Application: Self-Sufficiency Program

Example from Card and Hyslop (2005):

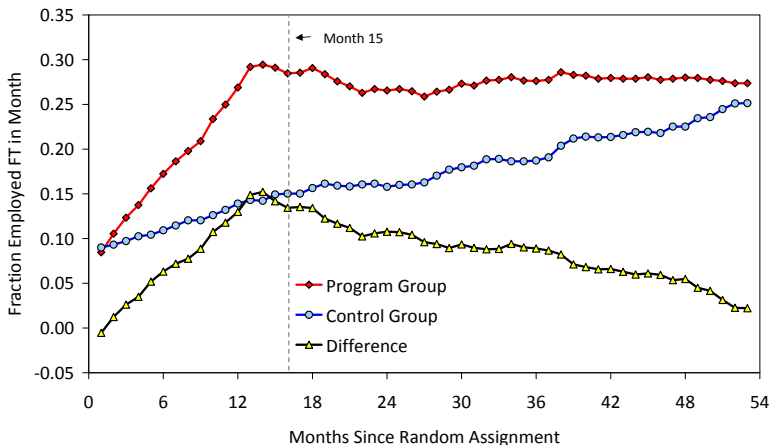
- Self Sufficiency Project (SSP) experiment, conducted in Canada, early 1990s
- Earnings subsidy (negative tax) for people who enter full-time (FT) work
- Subsidy = $0.5 \times (T - \text{monthly earnings})$ $T = \$2500/\text{mo}$ or $\$3083/\text{mo}$
- ONLY for the FT (30+ hrs/week) who earn $\geq$ the minimum wage:
 - subsidy lasts for 3 years;
 - have to start receiving subsidy within 12 mo. of random assignment.
- **Research Questions:** SSP subsidy = strong incentive to enter FT work, which is a “short-term establishment incentive”.
 - 1 can long-term welfare recipients find a job and leave welfare? (short-term establishment)
 - 2 can they maintain a FT job and be better off in the long-term? (long-term entitlement as they choose work over welfare)

Experiment:

- Welfare recipients with kids (95% females), 1+ years on welfare
- Randomly allocated to treatment group ($z = 1$) or control ($z = 0$)
- Series of surveys measure employment etc.

Application: Self-Sufficiency Program

Monthly FT Empl. Rates of SSP Control and Treatment Groups



- Outcome "Y": $IA15_i = 1$ if on welfare in month 15
- Endogenous variable of interest "D": $FT15_i = 1$ if the worker is employed full-time in month 15.
- Instrument "Z": random assignment to treatment group in the SSP experiment
- Causal model for receiving income assistance (IA) at month 15:

$$IA15_i = \beta_0 + \beta_1 FT15_i + u_i$$

- First stage model for FT empl at month 15:

$$FT15_i = \pi_0 + \pi_1 TREAT_i + \eta_i$$

- Reduced form model for IA at month 15:

$$IA15_i = \delta_0 + \delta_1 TREAT_i + \nu_i$$

- Underlying assumptions:

- ① Treatment is randomly assigned

$$(FT(0), FT(1)) \perp TREAT_i$$

- ② Exclusion restriction $Y(d, z) = Y(d)$: SSP only affects *IA* through full-time work? *SSP subsidy* $\Rightarrow$ *FT work* $\Rightarrow$ *leave IA* seems plausible.
- ③ Is “no defiers” satisfied? Likely since SSP would never cause you to stop working FT (unless you are really weird).

Summary of SSP Outcomes at Month 15

	Employed FT (1)	On Welfare (IA) (2)
Control Group (n=2718)	0.1472	0.8098
Treatment Group (n=2762)	0.2848	0.6687
Difference: T-C (std error)	0.1376 (0.011)	-0.1411 (0.012)
IV estimate:	-1.027 (0.082)	

- We know that there is only partial compliance with the subsidy incentive.

$$Pr(NT, mo.15) = 1 - 0.2848 = 0.7152$$

$$Pr(AT, mo.15) = 0.1472$$

$$Pr(C, mo.15) = 0.1376$$

- Note that these fractions are changing as the experiment goes on (x-axis = months since random assignment)

Q: who are the compliers? Are the “like” other people? Are they more or less advantaged?

Characterize Compilers

- X_i = HS graduation
- Apply goofy 2SLS to outcome “D*X” $\sim X_i \times FT15_i$

$$\begin{aligned}X_i FT15_i &= a + b \times FT15_i + u_i \\ FT15_i &= \pi_0 + \pi_1 Z_i + \nu_i\end{aligned}$$

$$\begin{aligned}b^{IV} &= \frac{\delta_1}{\pi_1} \text{ via reduced-form} \\ &= \frac{E[X_i FT15_i | Z_i = 1] - E[X_i FT15_i | Z_i = 0]}{E[FT15_i | Z_i = 1] - E[FT15_i | Z_i = 0]} \\ &= \frac{\text{cov}(X_i FT15_i, \widehat{FT15_i})}{\text{var}(\widehat{FT15_i})} \text{ via 2SLS}\end{aligned}$$

where predicted $\widehat{FT15_i} = \pi_0 + \pi_1 Z_i$. Note
 $\text{cov}(X_i FT15_i, \widehat{FT15_i}) = \pi_1 \times \text{cov}(X_i FT15_i, Z_i)$ and
 $\text{var}(\widehat{FT15_i}) = (\pi_1)^2 \times \text{var}(Z_i)$.

Characterize Compilers

Two-Stage Least Squares Estimation

Model d_hsggrad
Dependent Variable d_hsggrad

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	7.401383	7.401383	139.92	<.0001
Error	5478	289.7814	0.052899		
Corrected Total	5479	609.8392			

Root MSE	0.23000	R-Square	0.02491
Dependent Mean	0.12755	Adj R-Sq	0.02473
Coeff Var	180.31324		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variable Label
Intercept	1	0.011784	0.010269	1.15	0.2512	Intercept
EMFTM15	1	0.534926	0.045223	11.83	<.0001	Msurv:Employed full-time in mo15

Application: The Health Impacts of C-section

- Card, Fenizia, Silver: “The Health Impacts of Hospital Delivery Practices”
- Research Question: treatment practices vary widely across hospitals. In the context of childbirth, the authors assess impacts of the differences in delivery practices at childbirth on infant and maternal health
- Data: births in CA, with detailed information such as where patients live, where they deliver kids, delivery practices, health outcomes through records of ER visits and medical claims...
 - they classify hospitals based on c-section rates: H if a “high c-section” hospital, L otherwise
 - D_i = deliver at H hospital
 - y_i = c-section delivery; baby is readmitted to hospital after birth
- Empirical challenge: there is endogenous sorting in c-section (e.g., high risk, unobserved preferences...)

Application: The Health Impacts of C-section

- Quasi-experiment Design:
 - living closer to a H hospital than a L *exogenously* increases the chance that they choose a H hospital;
 - Z_i = relative distance of mom's home to H vs L hospital (continuous or indicator)
 - There are 2 types of compliers:
 - H -compliers: mothers who choose a H hospital because they live closer to H ;
 - $H\&C$ -compliers: mothers who choose a H hospital because they live closer to H and will choose c-section; if they live closer to a L they would not have used c-section.

First-stage: “H” or “C&H” Compliers

Assume: (1) random assign (2) exclusion restriction and (3) monotonicity.

deliver at H hospital: $H_i = \pi_0 + \pi_1 Z_i + \pi_X X_i + \epsilon_i$

deliver by C-section: $C_i = \gamma_0 + \gamma_1 Z_i + \gamma_X X_i + \eta_i$

Interpretations:

- π_1 : the fraction of H-compliers, who will choose a H hospital if they live closer to H. Note this includes people who would have chosen C-section at both (H, L) hospitals (“C” always takers), and people who were to choose c-section if they chose a H-hospital (C&H compliers).
- γ_1 : the fraction of “C&H” compliers. Why? Under monotonicity assumption, the only group who changes C_i are C&H compliers who choose a H hospital because they live closer to H, and who choose C-section because they are at H-hospitals.

First-stage: “H” or “C&H” Compliers

Table III: Estimated Effects of Relative Distance on Place, Mode, and Timing of Delivery

Outcome Variable	Mean	Instrument=Relative Distance to H Hospital Coefficients × 100		Instrument= Indicator for Closer to H Hospital Coefficients × 10	
		Baseline controls only	All controls	Baseline controls only	All controls
	(1)	(2)	(3)	(4)	(5)
Deliver at H Hospital	0.515	1.304 (0.135)	1.328 (0.134)	0.682 (0.094)	0.694 (0.091)
C-section Delivery	0.256	0.159 (0.028)	0.149 (0.027)	0.078 (0.021)	0.080 (0.020)
<i>Breakdown of C-Section Deliveries:</i>					
C-Section at H Hospital	0.149	0.401 (0.041)	0.403 (0.041)	0.208 (0.030)	0.213 (0.030)
C-Section at L Hospital	0.106	-0.243 (0.033)	-0.254 (0.030)	-0.130 (0.024)	-0.133 (0.022)
<i>Fractions of Complier Groups -- Moving 7 mi. closer to H hospital (col. 2-3) or closer to H hospital (col. 4-5)</i>					
P(H Complier)		0.091	0.093	0.068	0.069
P(C&H Complier)		0.011	0.010	0.008	0.008
P(H Complier & C Always-Taker)		0.017	0.018	0.013	0.013

IV-LATE for “H” Compliers

- Focusing on the H-hospital compliers, estimate the 2SLS model:

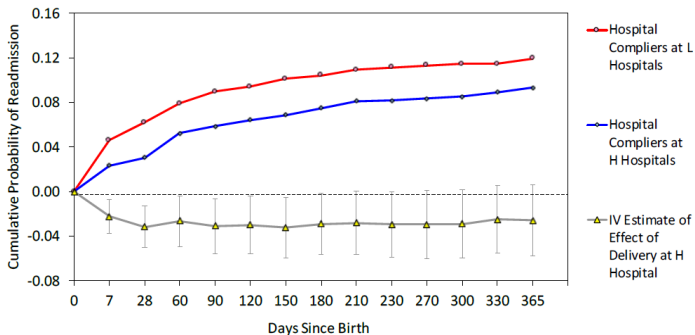
$$\begin{aligned}Y_i &= \beta_0 + \beta_1 H_i + \beta_X X_i + u_i \\&= \delta_0 + \delta_1 Z_i + \delta_X X_i + \nu_i \\ \Rightarrow \beta_1^{IV} &= \frac{\delta_1}{\pi_1}\end{aligned}$$

the IV estimate represents the LATE for H-compliers – the change in outcome y per additional delivery due to H hospital among the compliers.

- The authors find mixed health impacts: in the short run readmission rate is lower, but in the longer run ER visits are higher among the compliers.

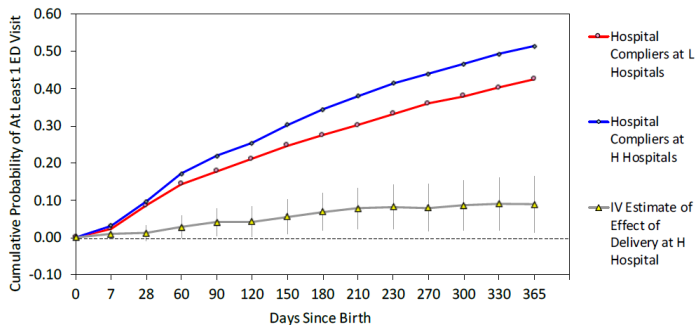
IV-LATE for “H” Compliers

a. Hospital Readmissions



IV-LATE for “H” Compliers

b. Emergency Department Visits



- Question: How did they estimate $E[Y_i(1)|Comp]$, $E[Y_i(0)|Comp]$?
Hint: goofy 2sls on $Y_i D_i$ and $Y_i(1 - D_i)$

Characteristics of Compliers

- The “relative distance” instrument is continuous: Z_i = distance to the closest Low-csection hospital relative to the closest H-hospital.
- But we can apply the goofy 2SLS to characterize the compliers:

$$\text{Causal Model: } X_i D_i = a + b D_i + e_i$$

$$\text{First stage: } D_i = \pi_0 + \pi_1 Z_i + \eta_i$$

$$b^{IV} = \frac{\text{cov}(X_i D_i, Z_i)}{\text{cov}(D_i, Z_i)} = \frac{E[X(Z_i - E[Z_i]) | D = 1]}{E[Z_i - E[Z_i] | D = 1]}$$

Table IV: Characteristics of Compliers

	Analysis Sample	Compliers		Ratio: C's/All (%)
<u>Socio-Economic Characteristics of Mother</u>				
<u>Race/Ethnicity</u>				
White	31.7%	25.9	(5.1)	81.6
Black	5.6%	0.7	(1.5)	12.4
Asian	17.6%	11.3	(4.8)	64.1
Hispanic	44.2%	62.2	(5.7)	140.7
<u>Education</u>				
High School or Less	41.2%	54.4	(4.7)	132.0
Some College	20.1%	12.9	(2.1)	64.5
BA or Higher	38.7%	32.6	(5.3)	84.4
Government Insurance	43.3%	62.4	(5.7)	144.2
<u>Predicted Probabilities of Infant ED Visit/Inpatient Admissions</u>				
Prob(ED Visit) > median	50.0%	67.8	(6.3)	135.5
Prob(Inpat. Adm) > median	50.0%	71.0	(6.4)	142.1

Characterize Compliers - Continuous Instrument

$$\begin{aligned} b^{IV} &= \frac{\text{cov}(X_i D_i, Z_i)}{\text{cov}(D_i, Z_i)} = \frac{E[X(Z_i - E[Z_i]) | D = 1]}{E[Z_i - E[Z_i] | D = 1]} \\ &= \int_{\underline{z}} E[X | \text{Comp}(\underline{z})] \times \omega(\underline{z}) d\underline{z} \end{aligned}$$

is weighted average of $E[X | \text{Comp}(\underline{z})]$ across complier groups, with weights $\int \omega(\underline{z}) d\underline{z} = 1$.

- Think of $\text{Comp}(\underline{z})$ as compliers who would have $D_i(z) = 1[Z_i \geq \underline{z}]$ – folks who would deliver at H-hospitals if the closest L-hospital is $\underline{z}$ miles further than the closest H.
- For always takers, $\underline{z} = -\infty$, and for never takers, $\underline{z} = \infty$.
- In the binary instrument case, there is only 1 group of compliers: $D_i(1) > D_i(0)$.

Characterize Compliers - Continuous Instrument

- Let's classify units by $\underline{z}$ (resistance to treatment) where $D_i(\underline{z}) = 1[Z_i \geq \underline{z}]$
- Denominator (applying random assign of Z): let $\tilde{Z}_i = Z_i - E[Z_i]$, and F represent the cdf of the $\underline{z}$,

$$\begin{aligned} E[\tilde{Z}|D = 1] &= E_{\underline{z}}[E[\tilde{Z}|D = 1, \underline{z}]] \text{ by the law of iterated exp.} \\ &= \int_{\underline{z}} E[\tilde{Z}|\underbrace{Z \geq \underline{z}}_{D=1}, \underline{z}]dF(\underline{z}) = \int_{\underline{z}} E[\tilde{Z}|Z \geq \underline{z}]dF(\underline{z}) \end{aligned}$$

- Numerator:

$$\begin{aligned} E[X\tilde{Z}|D = 1] &= \int_{\underline{z}} E[X\tilde{Z}|Z \geq \underline{z}, \underline{z}]dF(\underline{z}) \\ &= \int_{\underline{z}} E[X|\underline{z}] \times E[\tilde{Z}|Z \geq \underline{z}]dF(\underline{z}) \end{aligned}$$

by random assign X independent of Z

Characterize Compliers - Continuous Instrument

- So the ratio

$$\frac{E[X\tilde{Z}|D=1]}{E[\tilde{Z}|D=1]} = \int_{\underline{z}} E[X|\underline{z}] \times \underbrace{\frac{E[\tilde{Z}|Z \geq \underline{z}]f(\underline{z})}{\int_{\underline{z}} E[\tilde{Z}|Z \geq \underline{z}]f(\underline{z})d\underline{z}}}_{\text{weight } \omega(\underline{z})} d\underline{z}$$

where the weights sum to 1:

$$\int_{\underline{z}} \omega(\underline{z})d\underline{z} = \frac{\int_{\underline{z}} E[\tilde{Z}|Z \geq \underline{z}]f(\underline{z})d\underline{z}}{\int_{\underline{z}} E[\tilde{Z}|Z \geq \underline{z}]f(\underline{z})d\underline{z}} = 1.$$